

AI-Enhanced Career Guidance System for Personalized Career Pathways: A Semantic Multi-Modal Approach to Explainable Career Recommendation

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Abstract—Most Indian institutions' career counselling still resorts to periodic aptitude tests and one-off static questionnaires that rarely form part of an ongoing process, leading to an ever-growing divergence between the student's present knowledge pool and the current expectations of the job market. This paper puts forth an AI-Enabled Career Guidance System where the student profile is considered a living entity rather than a mere snap-shot, dynamically aggregating academic data, resume details, technical and soft-skill assessment data, course completions and certifications, interests, and behavioural signals into an aggregate evolving student representation. At the heart of our approach is our proposed Semantic Multi-Modal Career Alignment (SMMCA) model that fuses these disparate signals into a unified feature-vector for career characteristics. This vector is then scored against timely and granular labour market trends employing sentence-transformer based semantic embeddings, cosine similarity, and a LightGBM classifier trained on longitudinal student employment and career outcome data. We go beyond recommending a bare career label; instead, every recommendation is supplemented by: the specific strengths leading to the recommendation and the corresponding weaknesses hampering the fit; a predicted placement-readiness score; and, an adaptive learning path progressing from beginner to advanced stages via content-based filtering. We detail our underlying layered React/Spring Boot/PostgreSQL system architecture and ten core modules divided between student, mentor and administrator roles, embedded within a role-based workflow where mentors validate student recommendations against those proposed through intelligent automation. We conclude with comparative results versus existing resume-matching and course recommendation systems, an analysis of current shortcomings, namely 'cold-start' users and labour market data staleness, and a roadmap toward NCrf-enabled credit tracking and detailed analytics for placement cells.

Keywords: Career guidance, explainable AI, resume intelligence, semantic embeddings, skill-gap analysis, LightGBM, SHAP, recommendation systems, placement readiness, natural language processing, National Credit Framework

I. INTRODUCTION

It is one of the most significant life decisions a young adult will ever make, and yet in practice, how a person picks their first career hinges on nothing more than which career

path his/hers friend chose, an aptitude test given in tenth grade or advice from a relative that last visited the job market a decade prior [1],[2]. In colleges, career counselling cells are understaffed and often do nothing more than administering printed questionnaire and using spreadsheet for previous year placement statistics [1],[2]. As a consequence, students are systematically trained for careers that nobody is hiring for, and as is natural students develop career changes later or suffer from a lack of placement or feel the need to retrain from scratch as they become aware that they are either completely misfit or are simply incapable of finding relevant jobs [3].

Existing solutions focus on solving isolated sub-problems but don't bring them all together. Resume analyser scan extract useful keywords but don't take any action in advising what path should be taken, aptitude testing platform scan classify a user into different types but offer a vague connection between these type and current relevant jobs, and course recommendational engines just offer a limited static catalogue of relevant courses rather than analysing a user skill gaps to suggest targeted learning objectives [3], [4]. There are no holistic tools and as such, students feel that they are bombarded with useless and redundant information [4].

Here, we describe a prototype of an AI-powered career coaching system that intends to remedy this situation by continuously updating the state of a student as the primary subject of its actions rather than as static data to be fed to a system just once. In other words, whenever an action by student causes the state to update (resuming, a skill or Aptitude testing, adding another certification, uploading CGPA), it re-assesses state of a student relative to a live view of demand from recruiters and proposes corresponding appropriate actions. At its core is a proposed Semantic Multimodal Career Alignment (SMCA) Model which combines the various signals of data for each individual: Academic, Resume, Assessment, Interests and Personality. Then, it uses the Sentence-Transformer embeddings combined with LightGBM classifier to score each and every one of career categories and for each recommended career aSHAP[6], [7]explains why that option was chosen and where the student needs to venture to further boost compatibility score.

This paper proceeds as follows: Second part briefly discusses prior work regarding resume intelligence, Career Recommendation and Explainable AI. Third part elaborates on the problem statement and the motivation behind building this system. Fourth Part illustrates the

architecture and functional segments of our system and also discusses workflow between the segments. We then provide the complete detail explanation of all ten core segments and plan the system's decomposition over a small team of three people as the third part of our paper provides the system architecture as a result that the systems can be develop in two semester time span, as the remaining five parts of the paper include complete implementation of the system[11],[9].

II. LITERATURE SURVEY

Traditionally, career recommendation is regarded as classification task on a fixed set of features that are called aptitude features. Hand-crafted rule based counseling systems related test scores to career clusters by creating decision trees that are easy to audit but are brittle if a student's profile does not align with one of the predefined clusters [1]. More recently, collaborative-filtering and content-based recommenders, which compare a student's self-reported interests with the placement history of other students with similar interests, have been proposed as a method to suggest a career [7] as an ecommerce system suggests products for a customer to buy. These methods enhance static rule trees, but still rely on the quality and timeliness of the historical data they are trained on and do not typically make use of a student's resume's content.

For the resume side, a significant amount of work has been spent on extracting structured information from unstructured resume text, such as extracting skills, degrees and work history with reasonable accuracy across a variety of resume formats [8, 9] using named-entity recognition and section-classification models. With a Siamese BERT network, both resumes and job descriptions were embedded into a common semantic space, allowing for similarity search without requiring a full cross-encoder pass over every candidate pair, as achieved by Reimers and Gurevych's Sentence-BERT architecture [10]. This type of sentence-embedding model is used by most recent resume-to-job matching pipelines, such as the one picked in this project for matching student profiles with career descriptions.

Another strand of research has focused on skill-gap analysis as a set-difference problem – the skills that a target role requires versus the skills that a candidate has, and a ranking of the missing skills based on their prevalence in job postings, so that the learner can focus on learning the skills most needed to be the first to learn [11]. The basic framing is adopted in this project, but it is extended by integrating the gap analysis with the output of the resume parser and the LightGBM career classifier thus computing the gap against the specific career the model predicts rather than a generic industry average.

Gradient-boosted tree ensembles like LightGBM are commonly used in such tabular classification problems because they can cope with mixed categorical and numerical features and have proven to be efficient to train even on relatively small data sets found in institutions, which is essential for a system that needs to be retrainable on a specific college placement data set and not on internet-sized data sets [12]. In this system, the LightGBM model was chosen as the architecture of the most powerful neural network because its original paper, LightGBM,

showed that the histogram-based gradient boosting can be as accurate as the previous boosting frameworks while being trained significantly quicker.

Explainability has been a recurring issue in any system that affects a person's education or career path, as a black-box recommendation is susceptible to distrust and difficult to follow. Lundberg and Lee's SHAP framework provides a principled method to explain the contribution of each input feature to the output of a model, and to attribute, for a given student, a Data Scientist's recommendation to high scores in Python and SQL and low scores in communication skills [13]. Previous work in education-technology has demonstrated that a prediction followed by a feature-level explanation is significantly more likely to get students to follow recommended actions, and to increase their confidence in the prediction, than a score alone [14].

Course and learning-resource recommendation has already been explored as a content-based filtering problem, where a learner's skill gap is compared to a catalogue of courses labelled with the set of skills covered by the course, and the courses are ordered in terms of the amount of the remaining skill gap that the course would cover [15]. This project takes a content-based approach to the learning-roadmap generator, but instead of returning a flat list of courses ranked by the student, a raw list is not useful, as it does not provide information to the student about which course depends on which.

Last but not least, Indian studies consistently point to lack of alignment and slow feedback between what the curricula teach and what employers hire for as a structural contributor to underemployment among otherwise qualified graduates, and to the need to continually monitor student placement readiness based on data instead of a single pre-placement talk in the final semester of study [16]. The deficit the present system is designed to fill is lack of a single platform that brings together continuous multi-modal profiling, resume intelligence, explainable career classification, and adaptive learning-path generation in a role-based platform that connects students, mentors and institutional administrators.

III. PROBLEM STATEMENT

Many students choose or continue with a career because they have incomplete information, such as a single aptitude test taken years ago, unformalized guidance or knowledge of just a few familiar job titles. The digital tools in use today serve to address only isolated slices of the decision and have not been tightly coupled to achieve a single view of the student that is continuously updated, nor have they offered explanations for the choice of a particular career as the best fit. What is needed is a system that (i) creates and continually updates a consolidated profile, based on a student's academic record, resume, assessment responses, certifications and expressed interests; (ii) rates each profile against the current set of career categories using a semantic model, and not just a keyword match; (iii) provides explanations for each recommendation that can be understood by a student and a mentor; and (iv) makes the resulting skill gap an ordered, actionable learning roadmap. Formally the system should return a list of career recommendations with a

compatibility score, a feature-level explanation and an output of skill-gap and learning-path associated with that career recommendation ranked by the level of compatibility each recommendation has with the student's profile, resume, and assessment scores, and make the rankings reflect the changes that happen when any of the elements of the tuple changes.

IV. PROPOSED SYSTEM ARCHITECTURE

The platform can be divided into five functional stages: (1) Data collection in Student module, Mentor module and Admin module, (2) Pre-processing of data (handling missing data, cleaning text data, encoding feature data), (3) Resume Intelligence (NLP parser to extract skills, education, projects, and certifications from the uploaded resume), (4) Feature Representation (The proposed SMMCA model fuses all collected signals into a single Career feature vector), (5) Recommendation and explanation (The LightGBM classifier, Cosine-similarity matcher, SHAP explainer generate ranked Career suggestions, Skill-gap report, Placement-readiness score, and Learning roadmap). Each phase has a well documented REST interface that allows all 3 sub-teams (Frontend, Backend and AI) to develop and test their parts of the puzzle before they start to be integrated.

Fig. 1. System Pipeline Architecture

A. Authentication and Profile Management

Students, mentors and administrators log in using JWT tokens secured by Spring Security, and the role of the user dictates which dashboard and API surface is shown. The idea of the profile record is that it is additive, not a one-shot form, so that every time a student resumes is uploaded, every time they take an assessment, and every time they take a certification, they are adding to the profile instead of replacing it, which enables the recommendation engine described below to treat the profile as a continuous stream of information rather than a fixed state of affairs.

B. Resume Intelligence

The resumes uploaded are extracted by layout-sensitive text extraction methods along with a model learnt from the named entity recognition (NER) of skills, degrees, project

names and certification names from unstructured resumes text [8] [9]. In the entities extraction stage, extracted entities are normalized against a pre-maintained taxonomy of skills, before the feature-representation stage, so that synonymous entities, such as 'React.js' and 'ReactJS', are unified and mapped to a single skill node.

C. Career Based Semantic Multi-Modal Alignment (SMMCA)

SMMCA model is the main contribution of this system. The features are embedded or Normalised individually into a similar numeric range and concatenated together into a single vector for each student entry, resulting in seven groups of features per student entry. Likewise, Career Categories are built the same way as the first: from a carefully chosen sample of job descriptions for the specific career, all the necessary competencies, and typical educational requirements are built. The student vector and the free-text of the career-role descriptions are embedded into a common semantic space using a sentence-transformer model (all-MiniLM-L6-v2) and cosine similarity is calculated between the student vector and each career vector, similar to the standard formulation employed in semantic search systems [10, 17].

D. Career Prediction and Explanation

The cosine-similarity ranking is combined with a LightGBM classifier which learns non-linear relationships among features (e.g., a high CGPA is correlated with research-oriented jobs, while communication scores are correlated with client-facing jobs), that would otherwise not be captured by a purely similarity-based ranking [12]. Each recommended career is then fed through a SHAP explainer that breaks down the reasoning for the prediction to individual features of the input and returns the strongest features and weakest features, providing a clear, feature-level explanation for each of the recommended careers – as opposed to just a label – to provide the student and their mentor with a concrete explanation for every recommendation rather than an opaque one [13].

E. Skill-Gap Analysis and Learning Roadmap

The system calculates the difference between the skills that a top career needs and the skills listed in the student's profile, and returns the difference in a top ranked list. The skills that are missing are then compared to a catalog of courses and certifications using a content-based filtering approach, the resulting list of courses is then ranked in a sequence from beginner to advanced, where prerequisite skills are suggested before the subsequent ones that build upon them, and the sequence of suggested courses is presented as a roadmap of courses. Content based filtering for missing skills is then used to match the missing skills to a catalog of courses and certifications, and the resulting list of courses is presented as a sequence of courses from beginner to advanced where prerequisite skills are suggested before the next skills that build upon them [15].

F. Data Flow Summary

Each of the pipeline stages is described at the interface between the sub-teams responsible for the Frontend, Backend and AI by the input and output of each stage as summarized in Table I.

Stage	Input	Output
Profile & Assessment Intake	Academic records, resume, quiz responses	Structured student profile
Resume Parsing	Uploaded resume (PDF/DOCX)	Extracted skills, education, projects
Feature Representation	Structured profile + resume features	Unified career feature vector
Career Prediction	Career feature vector	Ranked career options
Skill Gap Analysis	Predicted career + current skills	Missing skill list
Explanation Generation	Prediction + feature vector	SHAP-based explanation report
Learning Path & Placement Score	Skill gap + assessments	Roadmap + readiness score

TABLE I. PIPELINE STAGE INPUT/OUTPUT CONTRACT

G. Scalability and Deployment Considerations

The platform is intended to be used by individual students on their own devices, and by institutional placement cells for bulk analytics, hence the backend is designed as stateless, horizontally scalable instances behind a load balancer with the AI recommendation service being an independent python microservice, so that model retraining and version updates do not require redeploying the core application. Relational profile, assessment and placement data is stored in PostgreSQL and the resume text embeddings are generated and stored in the cache to prevent running the SMMCA vector again on every dashboard refresh.

V. MODULE DEVELOPMENT, DETAILED DESIGN AND ROADMAP

The project is divided into two semesters. Semester one provides a minimum viable product for profile management, resume parsing, and a first iteration of the career-prediction engine (Modules 1-5); Semester two builds on the MVI with explainability, mentor workflows,

placement-readiness prediction, and institutional analytics (Modules 6-10).

A. Semester 7 – MVP Modules

Module 1 – Authentication and Student Onboarding, students will learn how to use Spring Security and JWT for secure student, mentor, and admin registration and login, have role-based dashboards, and will have a student, mentor, and admin profile creation screen, culminating in a working scoped-session login flow for each role.

Module 2 – Profile and Academic Record Management: Structured capture of academic records, CGPA, certifications and interests, with an indicator for profile completeness, to encourage students to fill in missing information or interests that could otherwise reduce the SMMCA vector.

Module 3 – Resume Upload and NLP Parsing: A resume-upload pipeline, which feeds into an NLP parser, that extracts entities such as skill, education, project, and certification, normalizing these entities against a shared taxonomy of skills (described in Section IV-B).

Module 4 – Aptitude and Skill Assessment Engine: An assessment module that can be configured with technical, soft-skill and aptitude questions, and stores the results of each assessment directly into the student's profile vector, which changes over time.

Module 5 – Career Prediction Engine (v1): Sentence transformer embedding + Cosine similarity ranking + Initial LightGBM classifier on the fused SMMCA vector, giving the student a ranked list of career suggestions on student dashboard.

B. Semester 8 – Advanced Feature Modules.

Module 6 – Explainable AI Layer: Integrating the SHAP explainer on the LightGBM's predictions - showing the strongest and the weakest parts of the model for each recommendation, and presenting them as a readable explanation panel, not as numbers.

Module 7 – Skill-Gap Analysis and Adaptive Learning Roadmap: Creation of the Skill Gap report against the highest predicted Career path and generation of a course and certification roadmap, from beginner to advanced, based on content-based filtering.

Module 8 – Placement Readiness Prediction: A composite scoring module for predicting placement readiness as a result of a combination of skill coverage, resume quality, certifications, assessment results and project count, that is updated automatically whenever the underlying profile changes, from 0 to 100.

Module 9 – Mentor Dashboard and Verification Workflow: Mentor facing screens to review assigned students' recommendations and schedule guidance sessions as well as to leave structured feedback that is also fed back into subsequent profile evaluations to complement rather than supplant mentoring.

Module 10 – Admin Analytics, Integration and Deployment: Institutional dashboards and summary reports of placement readiness and trends in skill gaps by cohorts, followed by system-wide integration testing, performance tuning and deployment in the cloud, resulting in a production representative system.

C. Team Responsibilities

Kabilan S – Backend and Integration Architect: Developed the Spring Boot service layer, designed the REST API and the PostgreSQL schema, implemented JWT for security, and designed the integration contracts between the frontend and the Python AI microservice.

Midhun G – AI NLP Engineer: Responsible for the resume parsing pipeline, the SMMCA feature-fusion implementation, the LightGBM career classifier, the SHAP explanation layer and the content-based course-recommendation engine.

Sughadev J P – Frontend and UX Lead: Responsible for React and Material UI student, mentor, admin dashboards, resume-upload and assessment-taking screens for both semesters, explanation and roadmap visualization screens for both semesters.

VI. ECOSYSTEM TECHNOLOGY STACK

React.js and Material UI are used for the frontend because of their component libraries and ability to fit the platform's three different user roles, with its dashboard heavy, form-driven interfaces. Both Axios and React Router are used to handle API communication and navigation between the student view, mentor view and admin view of a single page application.

Backend is powered by Spring Boot, Spring Security (used for authentication), Spring Data JPA (used for persistence) and Hibernate (used as a persistence framework) for its maturity in enterprise applications that require roles and with its ease of integration with a PostgreSQL database. Flyway takes care of schema migrations, ensuring that the database's structure changes in unison with the module roadmap outlined in Section V.

The AI layer is written in Python, with the following libraries used: Scikit-Learn and pandas for the preprocessing, Sentence Transformers for embedding generation, spaCy for the Named-Entity Recognition in the resume parser, LightGBM for classification of the career and SHAP for generation of explanations. The AI service is deployed as a standalone microservice, accessed by the Spring Boot back end service via REST, as shown in Section IV-G.

Resume and career-description embeddings are stored separately to be cached, and profiles, assessments and placement records are stored in a primary relational store of PostgreSQL, which are recalculated only once. Email services and file upload services are used for notifying; any external labour-market data is synced through a REST integration that can be configured; and the career-description dataset can be refreshed without code changes.

VII. IMPLEMENTATION DETAILS

A. Data Model

There are 5 tables in the Data Model. The following are the Students table, Resumes table, Assessments table, Careers table and Recommendations table. The Students table has information like the students id name, contact, academic record and credential hash. The Resumes table has the id, student id parsed skills parsed education and parsed projects. The Assessments table has the id, student id, assessment type, score and timestamp. The Careers table has the id, title, required skills, academic and description embedding. The Recommendations table has the id, student id, career id, compatibility score, SHAP explanation payload and generated timestamp. Tables for skills and courses are also available. The taxonomy of the skills are used in these tables. This implies that the Data Model will utilize the resumes parser skill vocabulary, the assessment engine and the course recommender. This taxonomy is adopted in the DAM in order to refer to the same skill. This is essential, with respect to the Students Data Model, the Resumes Data Model, the Assessments Data Model, the Careers Data Model and the Recommendations Data Model.

B. SMMCA Feature Fusion and Retrieval Formulation

The student profile is represented as a string of seven sub-vectors because it includes academic, resume, assessment, interest, personality, certification and preference features; this string is denoted as a single vector s . It's the same for career category c , which is based on curated roles data. Both s and each candidate ci are projected into a shared d -dimensional embedding space by the sentence-transformer model. Cosine similarity is then used to measure the compatibility between the two vectors, $\text{sim}(s, ci) = (s \cdot ci) / (\|s\| \cdot \|ci\|)$, where the inner product of the two vectors is $s \cdot ci$ and the L2 norm of the two vectors are $\|s\|$ and $\|ci\|$. In a similar fashion to semantic search and recommendation systems [10, 17], the top- k candidates of similarity scores are passed to the LightGBM classifier which re-ranks the candidates by exploiting learned non-linear feature interactions from historical placement data.

C. Explanation Generation

SHAP values are calculated for each of the input features of the fused vector s and the features are ordered by their signed contribution to the predicted class for the top-ranked career prediction. The positive contributors are surfaced as strengths to the student, e.g., high score in Python and SQL, and a number of projects in data, etc. for a recommendation to become a Data Scientist; whereas, the negative contributors are surfaced as weak areas that require attention, e.g., low score in communication skills, experience in cloud computing, etc. [13].

D. Continuous Profile Evolution

The fused vector s is re-calculated each time a student uploads a new resume, completes another assessment, or adds another certification to their profile, and the career suggestions, skill gap, and placement-readiness score always display the current version of the student's profile, and not a profile from an older session.



Fig. 2. Profile-to-Recommendation Sequence

VIII. Security, Privacy, AND DATA GOVERNANCE

Academic records are personally identifiable, assessment scores are sensitive and even the content of a resume is personally identifiable and sensitive information, and hence access control is not only enforced at the front end, but also at the API level. All traffic between the client and the back end is TLS-secured, and passwords are not stored as a reversible function, but are instead stored as a BCrypt hash.

Only the student can view their full profile, and only mentors who have been specifically designated to them can access it; the administrator can only view aggregated, cohort-level analytics, unless a specific verification workflow requires otherwise, where the administrator will be able to view the raw individual assessment answers. Each recommendation made to a student is recorded with a SHAP explanation, ensuring that both the student and any reviewing mentor can see why a particular career was recommended to address concerns raised in the broader literature on transparency in the use of AI decision support tools in education [14].

IX. TESTING AND VALIDATION STRATEGY

The following four Area specific testing procedures are conducted in Module 10. Automated unit and integration tests ensure that all backend endpoints are tested against their documented contract, including authentication failure scenarios, malformed resumes uploaded, and against a staging PostgreSQL database with synthetic student data.

At model level, the precision and recall of the resume parser is tested against a manually labelled sample resume, the accuracy and F1-score of the LightGBM classifier is tested against a held-out part of historical placements, and the stability of the SHAP explanations is tested by repeatedly running the model on the same profile of a resume.

End-to-end test covers the delay between the resume upload to the complete recommended resume including explanation and roadmap, to ensure the delay is acceptable for interactive use of the dashboard even in a concurrent multi-users load. Security tests exploit the paths in Section VIII for privilege escalation and token replay.

X. RESULTS AND DISCUSSION

The system is validated by a phased demonstration incorporating a resume upload, a career prediction, a corresponding SHAP explanation panel, skill gaps calculation relative to the predicted career and generation of a sequenced learning roadmap with placement-readiness score. Relative to only-counselor-based on aptitude tests, the platform requires less effort from a student, as they only need to submit a resume and complete a brief assessment, and the explanation layer provides both a student and a mentor with a tangible basis to accept, question, or reject a recommendation, similar to previous explainable-recommendation studies [13], [14].

Table II compares the proposed system with three different types of tools in prevalent use: static aptitude-test portals, generic resume screening systems, and course-recommendation chatbots. The proposed system is the only one that merges all of the above in a single pipeline, namely, continuous multi-modal profiling, resume intelligence, explainable output, and placement-readiness score.

System	Continuous Profiling	Resume Intelligence	Explainable Output	Placement Score
Static Aptitude Test Portals [1]	No	No	No	No
Generic Resume Screeners [8]	No	Partial	No	No
Course Recommender Chatbots [15]	Partial	No	No	No
Proposed System (SMMCA)	Yes	Yes	Yes	Yes

TABLE II. COMPARISON WITH EXISTING SYSTEMS

The mentor-verification workflow in Module 9 is expected to add particular value for placement cells handling large cohorts, since it removes the need for a mentor to manually cross-reference a student's resume against job requirements before every guidance session, letting the mentor instead review an already-computed compatibility score and explanation and spend the session

on judgment calls rather than data collection. Quantifying the resulting reduction in mentor time-per-student and the improvement in placement-conversion rate is left for a future longitudinal evaluation once the system is deployed across a full placement cycle, as discussed in Section XII.

XI. LIMITATIONS AND CHALLENGES

There are four main design shortcomings of the current design. First, because the LightGBM classifier is trained on historical placement data from a small number of institutions, its accuracy at other institutions or other categories of jobs or regions has not been tested, and a larger and continually updated labour-market dataset would be needed to generalize with confidence.

Second, the system also has a problem of cold starts, where students have little information in their profiles (no resume, few answer to the assessments, no certificate etc), with the resulting SMMCA vector being sparse and the resulting recommendation being less confident: a fallback recommendation strategy based on the student's academic stream will be implemented but not yet done.

Third, like with any machine-learning classifier, the LightGBM model could be misclassified or a feature might be spuriously correlated and appear in the SHAP explanation; therefore, the mentor-verification workflow in Module 9 is not optional and must be performed anytime a student wants to follow one of their recommendations.

Fourth, the current course-recommendation catalogue is curated manually and would require a scalable ingestion pipeline, probably on external learning-platform APIs, to keep up with the ever-expanding number of new courses and certifications in the fast-changing worlds of technology.

XII. Conclusions and future scope:

In this paper, we outlined the design of an AI-Enhanced Career Guidance System, which leverages a proposed Semantic Multi-Modal Career Alignment model for fusing together the academic, resume, assessment, and interest signals and present a continuously updated student representation, the LightGBM career classifier and a SHAP-based explanation layer which facilitates the transparency and actionability of every recommendation. In 2 semesters, a three-member team using the 10 modules can go from MVP with an authenticated profile management system to a complete explainable, mentor-checked recommendation system that provides institutional analytics.

Future work will involve testing the LightGBM classifier on a larger, multi-institution placement database, developing an automated pipeline to ingest the course/certification catalog and keep it up to date, but without manual curation, and extending the placement-readiness score with longitudinal outcome data when students who have used the system actually begin their careers.

Mentor feedback gathered during Module 9 is supposed to be used in the long-term to continuously fine-tune the ranking of course-recommendations and the classification

of the career; this feedback will be used to improve the recommendations as they are used, instead of them being static based on the accuracy of the initial training.

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